RIG-ETALE ALGEBRAS OVER A PRINCIPAL IDEAL ARE COMPLETIONS OF FINITE TYPE ALGEBRAS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof displays the concrete data (c, f_1..f_r, divisibility, inclusion) obtained from part (6) before the final invocation, so the argument is complete at the named-lemma level; the reference environment citing Elkik is statement-side metadata

Editorial note. The label lemma-approximate-principal names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Lemma lemma-equivalent-with-artin part (6): for a presentation B = A[x_1,...,x_r]^/J over (A, I) it produces c >= 0 and f_1,...,f_r in J such that the Jacobian determinant det(d f_j / d x_i) divides a^c in B and a^c J is contained in (f_1,...,f_r) + J^2
- Lemma lemma-approximate-presentation-rig-smooth (delivered, L48002): converts precisely such presentation data into a finite type A-algebra C with B isomorphic to C^

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at 2286; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAW: sources/stacks-restricted/pinned-original-source.tex, source lines 1638–1638. Prerequisite pointer invoked at 2299; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AJU: sources/stacks-restricted/pinned-original-source.tex, source lines 1810–1810. Prerequisite pointer invoked at 2294; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAQ: sources/stacks-restricted/pinned-original-source.tex, source lines 1167–1167. Prerequisite pointer invoked at 2302; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AK7: sources/stacks-restricted/pinned-original-source.tex, source lines 1294–1294. Prerequisite pointer invoked at 2304; it names

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a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a Noetherian ring and $I = (a)$ a principal ideal. Let B be an object of (0AL4) which is rig-étale over (A, I). Then there exists a finite type A-algebra C and an isomorphism $B \cong C^\wedge$.*

Proof. Choose a presentation $B = A[x_1, \ldots, x_r]^\wedge/J$. By Lemma 0AJU part (6) we can find $c \geq 0$ and $f_1, \ldots, f_r \in J$ such that $\det_{1 \leq i,j \leq r}(\partial f_j/\partial x_i)$ divides a^c in B and $a^c J \subset (f_1, \ldots, f_r) + J^2$. Hence Lemma 0GAW applies. This finishes the proof, but we'd like to point out that in this case the use of Lemma 0GAQ can be replaced by the much easier Lemma 0AK7. □